

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 10%

QUESTIONS:2

Total Marks:20

Descriptive Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.

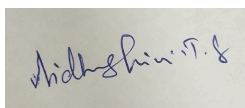
Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Code of Conduct:

I Vidhushini.T.S (192511061) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
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Understanding of Concepts	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	5
Explanation	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	5
Application	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	4
Organization	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	3
Accuracy	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	2
Clarity	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	1

Q1. Smart Home Automation System and Intelligent Agents:

A Smart Home Automation System is an Artificial Intelligence based system that automatically controls household devices such as lights, fans, air conditioners, heaters, security cameras, smart locks, and alarms. The system collects real-time information from various sensors such as motion sensors, temperature sensors, smoke detectors, door sensors, and time-based sensors. Based on this information, it makes intelligent decisions without requiring constant human intervention.

The main objectives of a smart home are to provide comfort, energy efficiency, security, and convenience. Since the environment changes continuously and user preferences also change over time, different types of intelligent agents are used to make better decisions. These agents perceive the environment through sensors, process the collected information, and perform suitable actions using actuators.

Types of Intelligent Agents:

1. Simple Reflex Agent:

A Simple Reflex Agent is the most basic type of intelligent agent. It makes decisions only by looking at the current input (percept) received from the sensors. It follows predefined condition-action rules, also known as "IF-THEN" rules. The agent does not store previous information, remember past events, or predict future situations.

For example, if a motion sensor detects movement inside a room, the agent immediately turns on the lights. If no motion is detected for five minutes, it switches the lights off. Similarly, if smoke is detected, the alarm is activated instantly.

Example Rules :

Condition (IF)	Action (THEN)
Motion detected	Turn ON lights
No motion for 5 minutes	Turn OFF lights
Smoke detected	Activate fire alarm
Door opened at midnight	Trigger security alarm

The biggest advantage of a Simple Reflex Agent is that it is very fast because it does not perform complex calculations. It is easy to design and requires very little memory. However, it has several limitations. Since it does not remember previous events, it cannot identify user habits or understand situations where complete information is unavailable. For example, if a person is sleeping and no movement is detected, the system may incorrectly switch off the lights.

Advantages:

- Fast response
- Easy to implement
- Low computational cost
- Suitable for simple automation tasks

Disadvantages:

- No memory of previous events
- Cannot learn user behaviour
- Cannot handle partially observable environments
- Not suitable for complex smart homes

2. Model-Based Reflex Agent:

A Model-Based Reflex Agent improves the capabilities of the Simple Reflex Agent by maintaining an internal model of the environment. It stores information about previous states and uses this memory to make better decisions.

Instead of depending only on the current sensor reading, the agent combines both current and past information to understand what is happening inside the house.

For example, suppose the bedroom motion sensor stops detecting movement because the user has fallen asleep. Unlike the Simple Reflex Agent, the Model-Based Agent remembers that someone entered the room recently and therefore does not switch off the lights or air conditioner immediately.

Similarly, if the outside temperature is dropping continuously, the system predicts that the indoor temperature will also decrease and starts the heater before the room becomes uncomfortable.

Internal Model Example :

Current Temperature= 22°C, Previous Temperature= 24°C, Outside Temperature= 18°C



Agent predicts room will become colder



Turns ON Heater automatically

The Model-Based Agent is also useful for security. If the homeowner usually leaves for work at 9:00 AM and returns at 6:00 PM, the system stores this information. If unexpected motion is detected during working hours, it identifies the activity as suspicious and activates the security system.

Advantages:

- Remembers previous events
- Handles partially observable environments
- Better decision making
- Suitable for dynamic environments

Disadvantages:

- More complex than Simple Reflex Agent
- Requires additional memory
- Needs continuous updating of the internal model

3. Goal-Based Agent:

A Goal-Based Agent goes one step further by making decisions based on specific goals rather than simply reacting to sensor inputs.

The agent first identifies the desired goal and then plans the best sequence of actions required to achieve it.

In a smart home automation system, several important goals may exist.

Goals of the Smart Home

- Maintain comfortable room temperature
- Reduce electricity consumption
- Improve home security
- Provide user convenience
- Prevent accidents like fire or gas leakage

Suppose the user wants to maintain the room temperature at 24°C. The Goal-Based Agent continuously compares the current temperature with the desired temperature. If the room becomes warmer, it starts the air conditioner. If it becomes colder, it activates the heater.

Similarly, if the goal is energy conservation, the system schedules appliances to operate during off-peak electricity hours. If the homeowner leaves the house, the agent automatically locks doors, turns off unnecessary appliances, activates security cameras, and switches lighting to security mode.

Because it plans before acting, the Goal-Based Agent produces better results than reflex agents. However, when multiple goals conflict with one another, it becomes difficult to decide which goal should receive higher priority.

Advantages:

- Makes intelligent plans
- Focuses on achieving objectives
- More flexible than reflex agents
- Suitable for complex decision-making

Disadvantages:

- Planning requires more computation
- Difficult to prioritize conflicting goals
- Slower than simple reflex systems

4. Utility-Based Agent:

A Utility-Based Agent is considered the most advanced intelligent agent because it not only achieves goals but also evaluates the quality of every possible solution using a utility function.

A utility function assigns numerical values to different outcomes. The agent selects the action that provides the maximum overall benefit.

In a smart home, several objectives often conflict. Running the air conditioner continuously provides excellent comfort but consumes more electricity. Turning it off saves electricity but reduces comfort. The Utility-Based Agent balances these conflicting objectives.

Utility Function Example:

Option	Comfort	Energy Saving	Security	Utility Score
AC ON continuously	High	Low	High	80
AC OFF	Low	High	High	65
AC at 24°C	High	Medium	High	95

The Utility-Based Agent chooses the third option because it provides maximum comfort while still saving energy.

The system can also learn user preferences over time. If it observes that the homeowner prefers dim lighting after 9 PM and a room temperature of 23°C during summer evenings, it automatically adjusts settings accordingly.

For security, instead of triggering an alarm for every movement, the Utility-Based Agent considers multiple sensor readings such as motion, camera images, time of day, occupancy schedules, and door status before making a decision. This reduces unnecessary false alarms.

Advantages:

- Produces optimal decisions
- Balances multiple objectives
- Learns user preferences
- Improves comfort and energy efficiency

Disadvantages:

- Complex design
- High computational requirements
- Requires accurate utility functions

Most Effective Agent (Hybrid Approach) :

- In real-world smart homes, a single intelligent agent cannot satisfy all requirements. Therefore, a hybrid approach is the most effective solution.
- The Simple Reflex Agent provides immediate responses such as switching on lights when motion is detected or activating alarms when smoke is sensed.
- The Model-Based Agent remembers previous events and user behaviour, allowing the system to make more accurate decisions.
- The Goal-Based Agent ensures that important objectives such as maintaining comfort, reducing electricity consumption, and protecting the home are achieved through proper planning.
- The Utility-Based Agent evaluates different alternatives and chooses the one that provides the maximum overall benefit by balancing comfort, energy efficiency, cost, and security.
- By combining all four types of agents, the smart home becomes faster, more intelligent, adaptive, and capable of learning user preferences over time.

Q2. Robot in an Unknown Maze using Uniform Cost Search (UCS) and Iterative Deepening Search (IDS)

A robot placed inside an unknown maze must find a path to the exit without having any prior information about the maze layout. The robot does not know where walls, obstacles, or the exit are located. It explores the maze using sensors, collects information about its surroundings, and decides which direction to move. Since the robot has no prior knowledge, it cannot use heuristic information. Therefore, uninformed search algorithms are suitable because they search for the solution without estimating the distance to the goal.

Two important uninformed search strategies are Uniform Cost Search (UCS) and Iterative Deepening Search (IDS). These algorithms help the robot systematically explore the maze and eventually find the exit.

Uninformed Search Strategies:

Uninformed search algorithms do not know where the goal is located. They only use the information available during the search process. These algorithms systematically explore the search space until a solution is found.

The two commonly used uninformed search algorithms are:

- Uniform Cost Search (UCS)
- Iterative Deepening Search (IDS)

1. Uniform Cost Search (UCS):

Uniform Cost Search is an uninformed search algorithm that always expands the node with the lowest cumulative path cost. Instead of selecting the path with the fewest steps, UCS selects the path with the minimum total cost.

This algorithm is useful when different paths have different movement costs. For example, some areas of the maze may contain rough terrain, slippery surfaces, stairs, or obstacles that require more battery power or time to cross.

The robot calculates the total cost of reaching every possible path and always chooses the least expensive one.

Working of Uniform Cost Search:

1. Start from the initial node.
2. Insert the start node into a priority queue.
3. Remove the node having the smallest path cost.
4. Expand all neighbouring nodes.
5. Update their cumulative costs.
6. Continue until the exit is reached.

Example:

Suppose the robot has two possible paths:

Path	Total Cost
Start $\rightarrow$ A $\rightarrow$ Exit	15
Start $\rightarrow$ B $\rightarrow$ C $\rightarrow$ Exit	10

Although the second path has more steps, UCS selects it because its total cost (10) is lower than 15.

UCS chooses:

Start $\rightarrow$ B $\rightarrow$ Exit

because the total cost is minimum.

Advantages of UCS:

- Guarantees the optimal solution.
- Finds the lowest-cost path.
- Works well when movement costs are different.
- Complete algorithm if all step costs are positive.

Disadvantages of UCS:

- Requires a large amount of memory.
- Slow for large search spaces.
- Stores many nodes in the priority queue.
- Time increases as maze complexity increases.

2. Iterative Deepening Search (IDS):

Iterative Deepening Search combines the advantages of Depth First Search (DFS) and Breadth First Search (BFS).

Instead of searching indefinitely like DFS, IDS performs repeated depth-limited searches.

The robot first searches up to depth 1. If the exit is not found, it restarts the search with depth 2, then depth 3, and continues increasing the depth limit until the goal is reached.

Since each search explores only a limited depth, the memory requirement remains very low.

Working of IDS:

1. Perform Depth Limited Search with limit = 1.
2. If goal is not found, increase depth limit.
3. Repeat the search with the new depth.
4. Continue increasing the limit until the exit is found.

Advantages of IDS:

- Requires very little memory.
- Guarantees finding the shallowest solution.
- Suitable for large search spaces.
- Combines benefits of DFS and BFS.

Disadvantages of IDS:

- Repeats the search many times.
- Root nodes are expanded repeatedly.
- Takes longer because of repeated computation.

Comparison between UCS and IDS

Feature	UCS	IDS
Uses Path Cost	Yes	No
Memory Requirement	High	Low
Finds Lowest Cost Path	Yes	No
Finds Shallowest Path	Not Always	Yes
Suitable for Equal Costs	Good	Excellent
Suitable for Different Costs	Best	Poor
Speed	Slower	Faster for small depth

Relation to Intelligent Agents:

The robot itself acts as an Intelligent Agent because it continuously performs the Perceive → Think → Act cycle.

Different intelligent agents behave differently inside the maze.

Simple Reflex Agent:

A Simple Reflex Agent reacts only to the current situation. If it sees an open path, it simply moves forward. Since it cannot remember previous locations, it may repeatedly enter the same dead end or move in circles. Therefore, it is not suitable for maze navigation.

Model-Based Agent:

A Model-Based Agent stores information about previously visited paths and builds an internal map of the maze. It avoids revisiting the same locations and makes better decisions because it remembers the environment. This significantly improves navigation efficiency.

Goal-Based Agent:

A Goal-Based Agent focuses on achieving a specific objective, which is reaching the maze exit. It evaluates different paths and selects actions that move it closer to the goal. Search algorithms such as UCS and IDS naturally support Goal-Based Agents because they systematically search for a path to the exit.

Utility-Based Agent:

A Utility-Based Agent goes beyond simply reaching the exit. It also considers factors such as travel time, battery consumption, risk, and safety. If multiple paths lead to the exit, the agent selects the one that provides the maximum overall benefit. For example, it may choose a slightly longer path if it saves battery power or avoids dangerous terrain.

Most Effective Combination:

If all paths have equal movement costs, a Goal-Based Agent combined with Iterative Deepening Search (IDS) is the best choice. IDS uses very little memory, guarantees finding the shallowest solution, and is highly suitable for robots with limited memory.

If movement costs vary, such as when different paths require different amounts of energy or time, a Model-Based Agent combined with Uniform Cost Search (UCS) is more effective. The Model-Based Agent remembers previously explored paths, while UCS guarantees the lowest-cost path to the exit.

In real-world robotic applications, the best solution is a hybrid approach that combines a Model-Based Agent with Goal-Based reasoning and Uniform Cost Search.

The internal model allows the robot to remember explored areas, the goal-based reasoning keeps the robot focused on reaching the exit, and UCS ensures that the chosen path has the minimum travel cost. This combination provides intelligent decision-making, efficient navigation, and optimal path planning.